Quiz 5

1) Linear production function : $Y = a \cdot N$.

Recall that a concave production function is : $Y = a \cdot N^{\alpha}$

Hence, a linear production function is the special case when $\alpha = 1$. For a concave production function, α should be between 0 and 1.

Now, we need to analyze the labor demand when $\alpha = 1$:

From $L = N(1+\tau(\theta))$ (see quiz 4) and $\max_{N} Y(N) - wL$,

FOC:

$$a - w(1+\tau(\theta)) = 0$$

$$a = w(1+\tau(\theta)) \quad (1)$$

Recall that, for a linear production function, $MPL = a$ $\left(\frac{\partial Y}{\partial N} = a \right)$

Recall the expression for bargained wage (from the Wage functions lectures)

$$W = (1-\beta)z + z \cdot MPL(1+r\cdot\theta) \quad (2)$$

$MPL = a$

Plug (2) into (1):

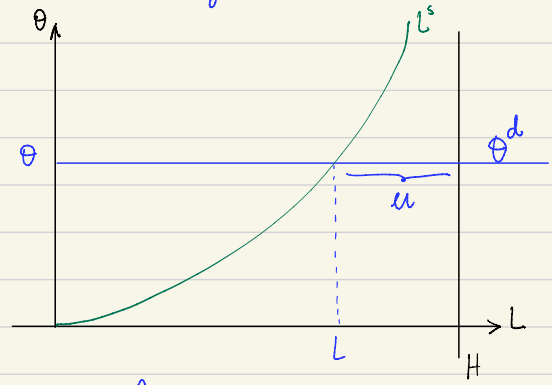
$$a = (1+\tau(\theta))\left[(1-\beta)z + z \cdot a(1+r\cdot\theta)\right]$$

$$\frac{a}{1+\tau(\theta)} = (1-\beta)z + z\,a(1+r\theta)$$

$$\frac{1}{r}\left[\frac{1}{z(1+\tau(\theta))} - \frac{(1-\beta)}{a} - 1\right] = \theta^{d}(a)$$

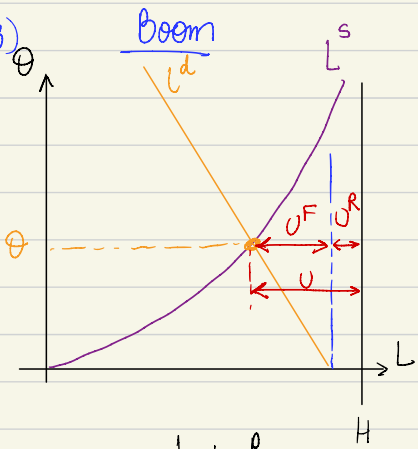
, which results in a perfectly elastic labor demand (horizontal $\theta^{d}(a)$)

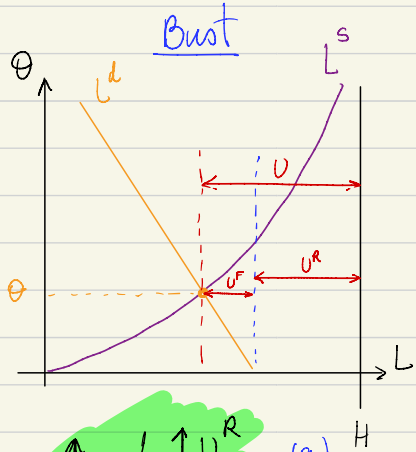
2) With a horizontal labor demand ($\theta^d(a)$):



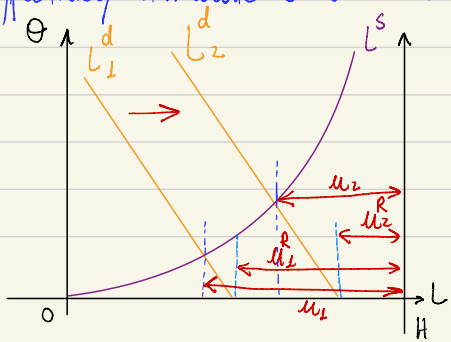
No, the job rationing model requires a concave production function, which allows for a downward-sloping $L^d(\theta, a)$

3)





$\downarrow U \begin{cases} \downarrow U^R \\ \uparrow U^F \end{cases}$

$\uparrow U \begin{cases} \uparrow U^R \\ \downarrow U^F \end{cases}$ (a)

4) In bad times, "U^R" is high. It happens when wages are too high relative to labor productivity. Thus, to reduce "U" during bad times, subsidizing wages is a way to effectively estimulate "L^d" when "a" is relatively low.